

Footwear Catalog Management in E-Commerce: A Study of Product Taxonomy, Attribute Standardization, and Category Hierarchy

Sanjana Ghosh

Senior Associate, Department of Data Tech - E-commerce Operations, MBA in Data Science, Kolkata,
West Bengal, India

Abstract

The rapid expansion of e-commerce platforms has increased the importance of effective catalog management for ensuring product discoverability, search relevance, and customer satisfaction. In the footwear industry, maintaining a structured product catalog is particularly challenging due to the wide variety of footwear types, complex category hierarchies, diverse product attributes, and large-scale inventory management requirements. This study investigates footwear catalog management through the development of a comprehensive framework focusing on product taxonomy, attribute standardization, and category hierarchy design.

The research utilizes a footwear taxonomy dataset consisting of approximately 129 category mappings organized across four hierarchical levels (L1-L4), an attribute repository containing approximately 1000+ attribute records, and a product image dataset representing various men's and women's footwear categories. The study examines taxonomy architecture, attribute governance, visual taxonomy validation, and catalog quality management practices within an e-commerce environment.

A structured methodology is proposed to analyze taxonomy consistency, attribute completeness, duplicate attribute detection, and image-based category verification. The research further develops a catalog management framework incorporating product onboarding workflows, taxonomy assignment rules, attribute validation mechanisms, quality control processes, and governance standards. Visual product images are utilized to validate footwear classifications and attribute mappings based on characteristics such as fastening type, toe shape, strap configuration, and footwear structure.

The findings indicate that standardized taxonomy structures, controlled attribute repositories, and image-assisted validation significantly improve catalog consistency, product classification accuracy, search relevance, and faceted navigation performance. The proposed framework provides a scalable approach for managing footwear catalogs and demonstrates the importance of structured product data in modern e-commerce systems. The study contributes to the fields of catalog management, product information management, and e-commerce data governance while offering practical insights for organizations seeking to improve catalog quality and operational efficiency.

Keywords: E-commerce Catalog Management, Footwear Taxonomy, Product Information Management (PIM), Attribute Standardization, Product Classification, Category Hierarchy, Catalog Quality Management, Visual Product Recognition, Product Attributes, Taxonomy Architecture, Catalog Governance, Search Relevance, Faceted Navigation, Computer Vision, E-commerce Data Management.



1.1 E-commerce Catalog Management

The rapid growth of e-commerce has transformed the retail industry by enabling customers to browse, compare, and purchase products through digital platforms. A fundamental component of every e-

commerce platform is the product catalog, which serves as the central repository of product information. Catalog management refers to the process of organizing, maintaining, and updating product data to ensure accurate representation, efficient product discovery, and enhanced customer experience.

In large online marketplaces, millions of products are categorized into hierarchical structures that enable customers to navigate products efficiently. Effective catalog management involves product classification, taxonomy development, attribute assignment, data validation, and quality control. A well-structured catalog improves search relevance, recommendation systems, faceted navigation, and overall user satisfaction.

Footwear is one of the most diverse product categories in e-commerce, encompassing various product types, styles, genders, and functional characteristics. Therefore, maintaining a consistent and scalable footwear catalog requires a robust taxonomy structure and standardized product information. Inconsistent categorization, duplicate classifications, and incomplete product attributes can negatively affect product visibility and customer purchasing decisions.

As online retailers continue to expand their product offerings, catalog management has become a strategic function that directly impacts operational efficiency, search performance, and sales conversion rates.

1.2 Product Information Management (PIM)

Product Information Management (PIM) is a systematic approach used to collect, manage, enrich, and distribute product information across multiple sales channels. A PIM framework ensures that product data remains accurate, consistent, and standardized throughout its lifecycle.

A typical PIM system stores various product-related attributes such as product name, category/product name, brand, type, color, gender, unisex, material, size, fastening, net quantity, season, occasion, warranty information, manufacturing information and product descriptions. By centralizing product information, organizations can reduce data inconsistencies and improve catalog quality.

In the footwear industry, PIM plays a significant role due to the complexity of product variations. Different footwear categories require different attribute sets. For example, sneakers may require attributes such as sole material and shoe type, while sandals may require sandal type and strap configuration. Effective management of such attributes ensures accurate product representation and improved discoverability.

Modern e-commerce platforms utilize PIM systems to support taxonomy management, attribute standardization, inventory synchronization, and marketplace integration. Furthermore, PIM facilitates better search engine optimization (SEO), recommendation accuracy, and customer decision-making by providing structured and comprehensive product information.

The integration of taxonomy frameworks with PIM systems enables organizations to maintain consistent category hierarchies while ensuring that all mandatory product attributes are accurately populated.

1.2 Footwear Industry Challenges

The footwear industry presents unique challenges in catalog management due to the extensive variety of product types, styles, materials, and consumer preferences. Products may be classified into numerous categories such as sneakers, loafers, pumps, boots, sandals, moccasins, sports shoes, etc., each having distinct characteristics and attribute requirements.

One of the major challenges is taxonomy inconsistency. Similar footwear products may be assigned to different categories because of naming variations, regional terminology, or manual classification errors. For example, categories such as "Mocassin" and "Moccasin" may represent the same product type but exist as separate classifications, leading to catalog fragmentation.

Another challenge involves attribute inconsistency and missing data. Product listings often contain incomplete or non-standardized attribute values, which negatively impact filtering, faceted navigation, and search relevance. Inaccurate attribute assignment can reduce customer confidence and increase product return rates.

Visual similarity among footwear products further complicates category assignment. Certain footwear styles share overlapping design characteristics, making manual classification difficult. As a result, image-based validation techniques are increasingly being explored to support category verification and catalog quality improvement.

Additionally, the continuous introduction of new styles, seasonal collections, and marketplace sellers creates challenges in maintaining taxonomy governance and data quality standards. Therefore, organizations require structured frameworks for taxonomy development, attribute standardization, and visual validation to ensure catalog consistency and scalability.

1.4 Research Objectives

The primary objective of this research is to examine footwear catalog management within an e-commerce environment through the analysis of product taxonomy structures, attribute repositories, and product images.

The specific objectives of the study are as follows:

1. To analyze the hierarchical structure of footwear taxonomy from Level 1 (L1) to Level 4 (L4).
2. To investigate the role of attribute standardization in maintaining catalog consistency and product data quality.
3. To identify taxonomy inconsistencies, duplicate classifications, and missing attribute values within the footwear catalog.
4. To develop a framework for footwear attribute governance and standardization.
5. To evaluate the use of product images for validating taxonomy assignments and category accuracy.
6. To propose a catalog management framework that integrates taxonomy management, attribute validation, and quality control mechanisms.
7. To assess the potential impact of taxonomy and attribute standardization on search relevance, faceted navigation, and overall catalog performance.

The findings of this study are expected to contribute to the fields of e-commerce catalog management, product information management, taxonomy engineering, and retail data quality by providing a structured framework for managing complex footwear product catalogs.

Chapter 2: Literature Review

Literature Review Framework



2.1 Product Taxonomy Systems

Product taxonomy refers to the hierarchical classification of products into structured categories and subcategories. In e-commerce environments, taxonomy systems serve as the foundation for organizing product information and enabling efficient product discovery. A well-designed taxonomy structure typically consists of multiple levels, ranging from broad categories to highly specific product types.

Researchers have emphasized the importance of taxonomy systems in improving catalog navigation, search performance, and inventory management. Hierarchical classification frameworks allow products to be grouped based on shared characteristics, facilitating easier browsing and filtering for customers. Common taxonomy structures follow a parent-child relationship where higher levels represent broader product groups and lower levels represent specialized categories.

In large-scale marketplaces, taxonomy design must balance comprehensiveness with usability. Excessively broad taxonomies may reduce search precision, whereas overly granular taxonomies can create maintenance challenges. Recent studies suggest that taxonomy governance and regular validation are essential for maintaining consistency as product catalogs expand.

For footwear products, taxonomy systems often incorporate multiple levels of classification, including gender, footwear category, footwear type, and style-specific classifications. Such structures support accurate product placement and enhance the customer shopping experience.

2.2 Catalog Quality Management

Catalog quality management focuses on maintaining the accuracy, completeness, consistency, and reliability of product information within e-commerce systems. High-quality catalogs contribute significantly to customer satisfaction, operational efficiency, and conversion rates.

Previous research has identified several dimensions of catalog quality, including data completeness, attribute accuracy, category consistency, and information timeliness. Poor catalog quality may lead to incorrect product representation, customer confusion, reduced search relevance, and increased return rates. Organizations employ various quality control mechanisms such as validation rules, automated audits, duplicate detection systems, and manual review processes to maintain catalog integrity. Data quality frameworks typically evaluate product records against predefined standards and identify discrepancies requiring correction.

Studies have demonstrated that improving catalog quality can directly enhance customer trust and facilitate better purchasing decisions. As e-commerce platforms continue to scale, catalog quality management has evolved from a support function to a strategic business requirement.

2.3 Attribute Standardization

Attribute standardization refers to the process of defining consistent attribute names, formats, and values across a product catalog. Standardized attributes enable accurate product comparison, filtering, analytics, and recommendation generation.

Literature in Product Information Management highlights attribute standardization as a critical component of catalog governance. Inconsistent attribute values often arise due to supplier-specific terminology, manual data entry errors, and regional naming variations. Such inconsistencies reduce data quality and negatively impact customer experience.

Researchers have proposed attribute dictionaries, controlled vocabularies, and standardized value repositories as mechanisms for ensuring consistency. Standardization frameworks generally define mandatory and optional attributes while establishing rules for accepted values.

Within footwear catalogs, common attributes include gender, size, color, material, fastening, sole type, upper material, lining material, product name, occasion, season, size type, manufacturing info, warranty information, and footwear style. Standardizing these attributes improves faceted navigation, search precision, and product discoverability. Furthermore, attribute standardization supports interoperability between Product Information Management systems and marketplace platforms.

Recent studies indicate that organizations implementing robust attribute governance frameworks achieve higher catalog consistency and reduced operational errors.

2.4 Product Classification Models

Product classification is the process of assigning products to appropriate categories based on their characteristics and attributes. Traditional classification approaches relied heavily on manual categorization performed by catalog specialists. However, the growth of large-scale e-commerce platforms has necessitated the adoption of automated classification models.

Machine learning and data-driven classification techniques have become increasingly popular in recent years. Classification models typically utilize structured product attributes, textual descriptions, and historical catalog data to predict appropriate category assignments. Common approaches include decision trees, support vector machines, random forests, and deep learning-based classifiers.

Research has demonstrated that automated classification systems can significantly reduce manual effort while improving categorization consistency. Nevertheless, classification accuracy remains dependent on the quality of input data and taxonomy design.

Hybrid approaches that combine rule-based systems with machine learning algorithms have shown promising results in complex product domains such as fashion and footwear. These approaches leverage both expert-defined taxonomy rules and predictive modeling techniques to improve classification performance.

In footwear catalogs, classification models are particularly useful for assigning products to categories such as sneakers, loafers, sandals, boots, sports shoes, etc., while minimizing human error.

2.5 Visual Product Recognition

Visual product recognition refers to the use of image analysis techniques to identify, classify, and validate products based on their visual characteristics. Advances in computer vision and deep learning have significantly expanded the application of visual recognition technologies within e-commerce.

Convolutional Neural Networks (CNNs) and image classification models are commonly used to analyze product images and detect distinguishing visual features. These technologies enable automated identification of product categories, styles, colors, and design elements.

Research suggests that visual recognition can complement traditional text-based classification systems by providing an additional validation layer. Image-based verification is particularly useful when product descriptions are incomplete, inaccurate, or inconsistent.

In the footwear domain, visual recognition systems can identify key characteristics such as sole design, strap configuration, heel structure, closure mechanisms, and overall silhouette. These visual features assist in differentiating among product types including loafers, pumps, sneakers, gladiators, and boots.

Recent studies have highlighted the growing role of image-based validation in improving catalog quality and reducing category assignment errors. As image datasets become more readily available, visual recognition is increasingly integrated into catalog management workflows.

2.6 Search and Recommendation Systems

Search and recommendation systems are essential components of modern e-commerce platforms. These systems help customers discover relevant products by analyzing product data, user behavior, and contextual information.

Search systems rely heavily on structured catalog information, including taxonomy hierarchies and standardized attributes. Accurate categorization and complete product attributes improve keyword

matching, filtering capabilities, and ranking algorithms. Poorly organized catalogs often result in irrelevant search results and reduced customer engagement.

Recommendation systems utilize various techniques such as collaborative filtering, content-based filtering, and hybrid recommendation models. Product taxonomy and attribute information play a crucial role in generating meaningful recommendations, particularly for new products with limited user interaction data.

Studies have shown that standardized product information significantly improves recommendation quality and personalization accuracy. Furthermore, faceted navigation systems depend on consistent attribute structures to enable efficient product filtering.

Within footwear e-commerce, search and recommendation systems assist customers in discovering products based on style preferences, footwear type, size availability, color options, and intended usage. Consequently, taxonomy quality and attribute standardization directly influence the effectiveness of these systems.

The literature collectively indicates that effective taxonomy management, attribute standardization, visual validation, and classification frameworks contribute substantially to improved search relevance, recommendation accuracy, and overall catalog performance.

Chapter 3: Research Methodology



3.1 Dataset Description

This study adopts a data-driven approach to analyze footwear catalog management in an e-commerce environment. The research utilizes three primary datasets: a footwear taxonomy dataset, an attribute repository, and a footwear image dataset. These datasets collectively represent the structural, descriptive, and visual dimensions of footwear catalog management.

The taxonomy dataset contains hierarchical product classifications organized across multiple levels. The attribute repository contains product attributes and their corresponding standardized values. The image

dataset consists of footwear images representing various product categories and styles for visual validation purposes.

The datasets were collected from an operational e-commerce catalog environment and were used to examine taxonomy consistency, attribute standardization, and category validation mechanisms. The study focuses on both men's and women's footwear categories, covering a broad range of footwear types including sandals, shoes, boots, sports footwear, and ethnic footwear.

The overall research methodology integrates taxonomy analysis, attribute quality assessment, and image-based validation to develop a comprehensive footwear catalog management framework.

Dataset Summary

Dataset	Description
L1-L4 Taxonomy Dataset	Hierarchical footwear classification structure
Attribute Repository	Product attributes and standardized values
Image Dataset	Men's and women's footwear images
Product Scope	Footwear
Gender Coverage	Men and Women
Analysis Type	Taxonomy, Attributes, and Visual Validation

Dataset	Description	Access Link
L1-L4 Taxonomy Dataset	Footwear category hierarchy	Footwear L1 to L4
Attribute Repository	Attribute names and LOVs	Footwear Attributes
Image Dataset	Footwear image collection	Footwear Visual Guide

The datasets used in this study are available through the links provided in the Table above. These datasets were utilized for taxonomy analysis, attribute standardization assessment, and visual taxonomy validation.

3.2 L1-L4 Taxonomy Dataset

The footwear taxonomy dataset forms the structural foundation of the study. The taxonomy is organized into four hierarchical levels (L1-L4), where each level provides increasing specificity in product classification.

Taxonomy Structure

L1 (Department Level):

- Footwear

L2 (Gender Level):

- Men
- Women
- Kids
- Infants
- Footwear Accessories

L3 (Category Level):

- Boys Casual

- Boys Sports
- Boys School
- Boys Ethnic
- Girls Casual
- Girls Sports
- Girls School
- Girls Ethnic
- Formal
- Casual
- Sports
- Ethnic
- Safety
- Boys Fashion
- Girls Fashion
- Shoe Accessories
- Shoe Care

L4 (Product Type Level):

- Sneaker
- Loafer
- Derby
- Oxford
- Gladiator
- Pump
- Chelsea Boot
- Biker Boot
- Sandal Types
- Various Specialized Footwear Types

The taxonomy dataset consists of 129 unique category mappings that establish relationships between different classification levels.

The taxonomy dataset was analyzed to identify:

- Category hierarchy consistency
- Duplicate classifications
- Naming standardization opportunities
- Category coverage
- Taxonomy depth and breadth

Special attention was given to identifying duplicate category names and inconsistent product classifications that could affect catalog quality and search relevance.

3.3 Attribute Repository

The attribute repository contains detailed product characteristics used to describe footwear products within the catalog. Attributes play a critical role in product discovery, faceted navigation, recommendation systems, and catalog governance.

The repository contains approximately 1000+ attribute records including attribute names and their associated List of Values (LOV).

Examples of Key Attributes

Attribute Name	Purpose
Gender	Product segmentation
Foot Size	Size specification
Sandal Type	Category-specific classification
Shoe/Boot Type	Product type identification
Fastening	Closure mechanism
Sole Material	Product construction
Warranty Type	Product assurance
Shoe Care Type	Maintenance information

The attribute repository was evaluated using the following dimensions:

Attribute Completeness

Assessment of mandatory and optional attribute population.

Attribute Consistency

Evaluation of uniformity in attribute naming and value representation.

Attribute Standardization

Identification of non-standard values and development of normalization rules.

Duplicate Detection

Detection of redundant attributes and overlapping value definitions.

Examples identified within the repository include variations such as:

- Mocassin vs Moccasin
- D'orsay vs D'Orsay
- Slide vs Sliders

These inconsistencies were analyzed to establish standard attribute governance practices.

3.4 Image Dataset

The image dataset consists of footwear product images representing both men's and women's categories.

The images were used to support visual taxonomy validation and category verification.

The dataset includes images from the following footwear groups:

Sandal Types

- Ankle Strap
- Back Strap
- Cross Strap
- Gladiator
- Fisherman
- Floater
- Huarache
- Jelly
- Jutti
- Loafer
- Mule
- Slide

- Sling Back
- Thong
- Toe Ring
- T-Strap
- V Strap

Shoe and Boot Types

- Sneaker
- Loafer
- Derby
- Oxford
- Brogue
- Chelsea
- Chukka
- Biker Boot
- Cowboy Boot
- Gumboot
- School Shoes
- Sport Shoes
- Snow Boot
- Slip-On
- Boat Shoes
- Canvas Shoes
- Casual Shoes
- Moccasin
- Monk Shoes

The image dataset was utilized for:

- Category verification
- Visual similarity assessment
- Product-type recognition
- Taxonomy validation
- Attribute-to-image mapping

Visual characteristics such as sole design, strap arrangement, closure type, heel structure, boot height, and silhouette were examined to determine whether product images aligned with their assigned taxonomy categories.

3.5 Analytical Framework

The analytical framework developed in this study integrates taxonomy analysis, attribute standardization, and image validation to evaluate footwear catalog quality.

Step 1: Taxonomy Analysis

The L1-L4 hierarchy was examined to identify:

- Structural consistency
- Category relationships
- Duplicate classifications
- Taxonomy completeness

Step 2: Attribute Assessment

The attribute repository was analyzed for:

- Mandatory attribute coverage
- Missing values
- Standardization opportunities
- Duplicate attributes
- Value consistency

Step 3: Image Validation

Footwear images were reviewed to verify:

- Category assignment accuracy
- Visual consistency
- Product type recognition
- Attribute correspondence

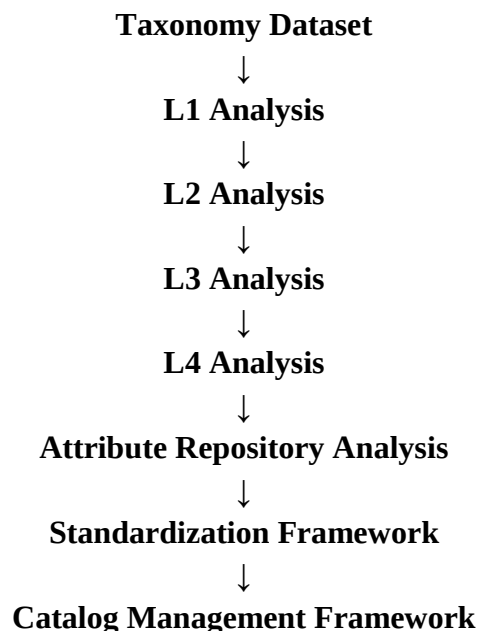
Step 4: Catalog Quality Evaluation

Results from taxonomy, attribute, and image analyses were combined to assess overall catalog quality.

Step 5: Framework Development

Based on the findings, a footwear catalog management framework was proposed to support:

- Product onboarding
- Taxonomy assignment
- Attribute validation
- Quality monitoring
- Catalog governance

Research Workflow

The proposed methodology provides a systematic approach for analyzing footwear taxonomy structures and attribute repositories, enabling the development of standardization and governance frameworks that enhance catalog consistency, data quality, and catalog management efficiency in e-commerce environments.

Chapter 4: Footwear Taxonomy Architecture

4.1 L1 Structure

The Level 1 (L1) structure represents the highest level of classification within the footwear catalog. It serves as the primary department under which all footwear products are organized. The purpose of the L1 structure is to establish a broad product grouping that supports catalog navigation, inventory management, and category governance.

In the present study, the L1 classification consists of a single department:

L1 Category

Level	Category
L1	Footwear

The Footwear department encompasses all footwear-related products available within the catalog and acts as the root node of the taxonomy hierarchy. All subsequent classification levels originate from this category.

A well-defined L1 structure simplifies catalog administration and provides a foundation for scalable taxonomy development. Since all products belong to a single footwear domain, the L1 structure remains highly stable and requires minimal modification over time.

4.2 L2 Structure

The Level 2 (L2) structure segments products based on gender-specific classification. Gender segmentation is widely adopted in footwear retail because product styles, sizing systems, and customer preferences differ significantly across demographic groups.

L2 Categories

Level	Category
L2	Men
L2	Women
L2	Kids
L2	Infants
L2	Footwear Accessories

The separation of products into Men's and Women's categories enables more accurate product discovery and supports personalized shopping experiences.

Benefits of L2 classification include:

- Improved customer navigation
- Better search refinement
- Gender-specific recommendations
- Enhanced merchandising strategies
- Accurate attribute assignment

The L2 layer acts as a bridge between the broad footwear department and more specialized footwear categories.

4.3 L3 Structure

The Level 3 (L3) structure introduces functional and category-specific groupings of footwear products. These categories represent major footwear segments that share common characteristics and usage patterns.

Some representative L3 Categories

L3 Category	Description
Sandals	Open footwear styles with straps
Shoes	Closed footwear styles
Boots	High-ankle or extended shaft footwear
Casual Footwear	Everyday usage footwear
Sports Footwear	Performance-oriented footwear
Ethnic Footwear	Traditional and cultural footwear

The L3 structure provides a logical grouping mechanism that reduces catalog complexity while preserving category-specific distinctions.

For example:

- Sandals contain open-foot designs.
- Shoes contain formal and casual closed designs.
- Boots contain ankle-length and calf-length footwear.
- Sports Footwear contains activity-based products.

The L3 layer plays a critical role in supporting faceted navigation and category-based merchandising strategies.

4.4 L4 Structure

Level 4 (L4) represents the most granular classification layer within the footwear taxonomy. This level contains individual footwear types that customers directly interact with during browsing and product selection.

The dataset contains approximately 129 unique L4 classifications.

Examples of Sandal-Type L4 Categories

Sandal Type
Ankle Strap
Back Strap
Cross Strap
Fisherman
Floater
Gladiator
Huarache
Jelly
Jutti
Loafer
Mule
Open Toe
Peeptoe
Slide

Sandal Type
Sling Back
Thong
Toe Ring
T-Strap
V Strap

Examples of Shoe/Boot-Type L4 Categories

Shoe/Boot Type
Sneaker
Loafer
Derby
Oxford
Brogue
Chelsea
Chukka
Biker Boot
Cowboy Boot
Boat Shoe
Canvas Shoe
Casual Shoe
Gumboot
Monk
Moccasin
Mojri
Pump
School Shoes
Slip-On
Snow Boot
Sports Shoes

The L4 level provides maximum specificity and directly influences:

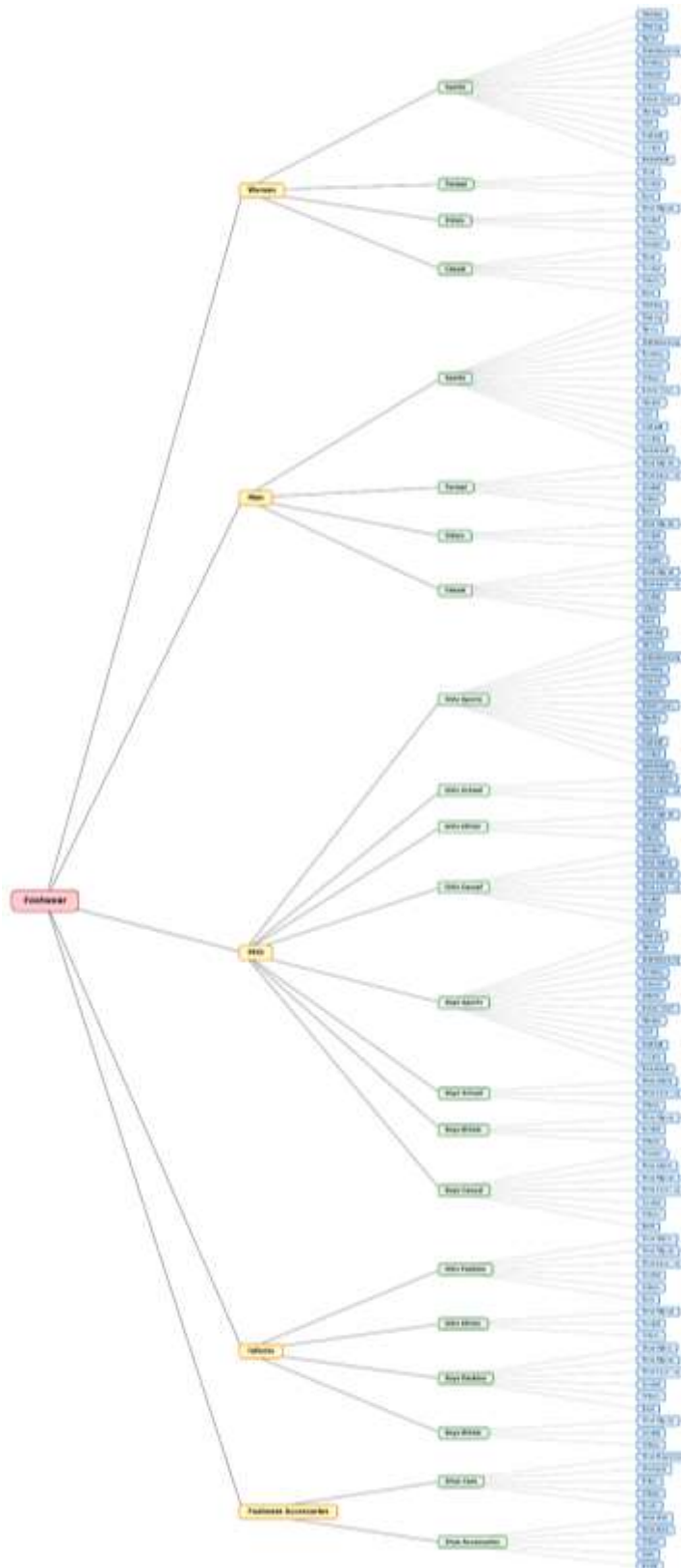
- Product search accuracy
- Faceted filtering
- Recommendation quality
- Catalog consistency
- Customer decision-making

Granular categorization enables users to locate desired products more efficiently while supporting accurate inventory organization.

4.5 Taxonomy Tree Visualization

The footwear taxonomy follows a hierarchical parent-child structure extending from L1 to L4.

Taxonomy Tree



The hierarchical structure demonstrates how broad product categories progressively narrow into highly specific footwear types.

4.6 Taxonomy Consistency Analysis

Taxonomy consistency is essential for maintaining catalog quality and ensuring accurate product classification. The footwear taxonomy dataset was analyzed to identify inconsistencies, duplicate categories, and naming variations.

Identified Taxonomy Issues

Duplicate Naming Variations

Variant 1	Variant 2
Mocassin	Moccasin
D'orsay	D'Orsay
Casual Shoes	Casual Shoe
Sports Shoes	Sport Shoes

Such variations can lead to fragmented product groups and inconsistent search results.

Category Overlap

Certain footwear types may appear across multiple category paths due to ambiguous classification rules. Examples include:

- Mule
- Jutti
- Jelly
- Slide

These categories may be classified as either sandals or shoes depending on marketplace conventions.

Missing Governance Standards

Without standardized taxonomy rules, different catalog operators may classify similar products differently, resulting in inconsistent customer experiences.

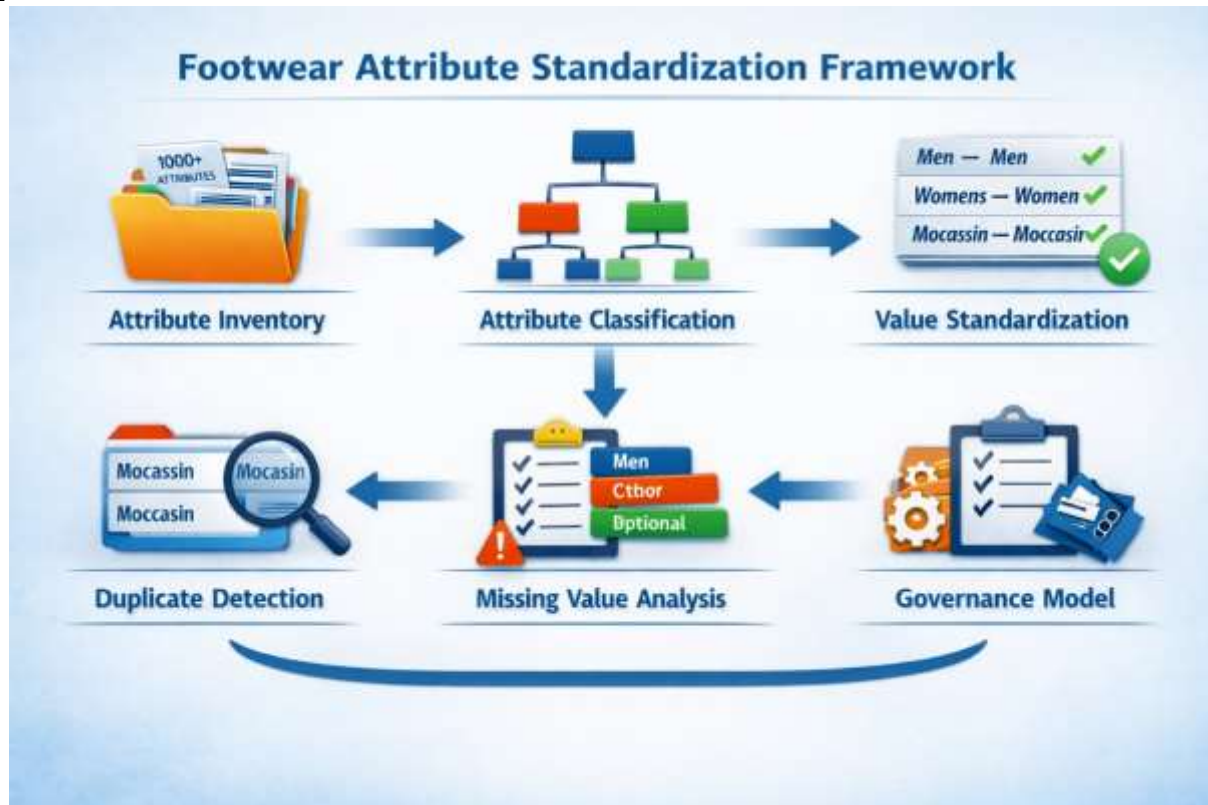
Recommended Standardization Framework

To improve taxonomy consistency, the following measures are recommended:

1. Establish a centralized taxonomy dictionary.
2. Define approved category names.
3. Eliminate duplicate category variants.
4. Implement automated category validation rules.
5. Conduct periodic taxonomy audits.
6. Align image-based validation with category assignments.

A consistent taxonomy structure improves search relevance, navigation efficiency, recommendation quality, and overall catalog governance. The analysis demonstrates that structured taxonomy management is critical for maintaining high-quality footwear catalogs in large-scale e-commerce environments.

Chapter 5: Footwear Attribute Standardization Framework



5.1 Attribute Inventory

Attributes serve as the descriptive elements that define the characteristics of a product within an e-commerce catalog. While taxonomy determines where a product belongs, attributes describe the product in detail and enable customers to compare, filter, and evaluate products effectively.

The footwear attribute repository analyzed in this study contains approximately 1000+ attribute records consisting of attribute names and their corresponding List of Values (LOVs). These attributes support product onboarding, search functionality, faceted navigation, recommendation systems, and catalog governance.

Major Attribute Categories

Product Identification Attributes

Attribute
Product Name
Brand
Gender
SKU
Product Type

Physical Attributes

Attribute
Foot Size
Color

Attribute
Material
Sole Material
Heel Height

Functional Attributes

Attribute
Fastening
Shoe Care Type
Warranty Type
Occasion

Category-Specific Attributes

Attribute
Sandal Type
Shoe Type
Boot Type
Tip Shape

The attribute inventory forms the foundation of catalog quality management by ensuring that products are consistently described across all categories.

5.2 Mandatory vs Optional Attributes

Not all attributes contribute equally to product discovery and catalog quality. Therefore, attributes are generally categorized as either mandatory or optional based on their business importance and customer relevance.

Mandatory Attributes

Mandatory attributes are essential for product publication and search functionality. Products lacking these attributes may experience reduced visibility or may not be eligible for listing.

Examples

Mandatory Attribute	Purpose
Gender	Category Assignment
Foot Size	Purchase Decision
Sandal/Shoe/Boot Type	Product Classification
Brand	Product Identification
Color	Filtering & Search
Product Name	Listing Visibility

Mandatory attributes ensure that products can be accurately categorized and discovered by customers.

Optional Attributes

Optional attributes provide supplementary information that enhances product descriptions but may not be required for publication.

Examples

Optional Attribute	Purpose
Warranty Type	Additional Product Information
Shoe Care Type	Maintenance Guidance
Style Name	Marketing Information
Collection Name	Product Segmentation

Although optional attributes are not compulsory, they improve customer understanding and contribute to a richer shopping experience.

5.3 Attribute Value Standardization

Attribute value standardization is the process of ensuring that attribute values follow consistent naming conventions and formatting rules throughout the catalog.

In large e-commerce environments, suppliers and catalog operators often use different terminologies to represent identical product characteristics. Such inconsistencies create fragmented data and negatively impact search performance.

Examples of Non-Standard Values

Non-Standard Values	Standardized Value
Men, Mens, Male	Men
Women, Womens, Female	Women
Mocassin, Moccasins	Moccasin
D'orsay, D'Orsay	D'Orsay
Sport Shoes, Sports Shoes	Sports Shoes

Standardization Rules

1. Use a single approved value for each concept.
2. Eliminate spelling variations.
3. Maintain consistent capitalization.
4. Remove duplicate synonyms.
5. Follow predefined LOV repositories.

Benefits of Standardization

- Improved search relevance
- Better faceted navigation
- Enhanced recommendation quality
- Consistent reporting and analytics
- Reduced catalog maintenance effort

Attribute standardization serves as a critical component of overall catalog governance.

5.4 Duplicate Attribute Detection

Duplicate attributes occur when multiple attributes or values represent the same product characteristic. Duplicate records create confusion for catalog operators and reduce data consistency.

Types of Duplication

Naming Duplication

Attribute Variant A	Attribute Variant B
Mocassin	Moccasin
D'orsay	D'Orsay
Sports Shoes	Sport Shoes

Semantic Duplication

Different terms may describe the same concept.

Example:

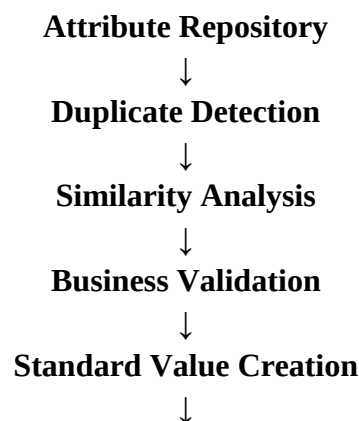
Value A	Value B
Slip-On	Slip On
Casual Shoes	Casual Shoe

Detection Approach

The study utilized the following duplicate detection techniques:

- Exact text matching
- Case-insensitive comparison
- Similarity-based matching
- Manual validation
- Controlled vocabulary review

Duplicate Resolution Framework



Repository Update

Removing duplicates contributes to higher catalog consistency and reduces classification ambiguity.

5.5 Missing Value Analysis

Missing attribute values represent one of the most common catalog quality issues in e-commerce environments. Incomplete product information reduces search visibility and negatively affects customer decision-making.

Causes of Missing Values

1. Supplier data omissions
2. Incomplete onboarding processes
3. Legacy catalog migration issues
4. Attribute mapping errors

5. Manual entry mistakes

Impact of Missing Values

Catalog Function	Impact
Search	Reduced relevance
Filters	Incomplete navigation
Recommendations	Lower accuracy
Product Comparison	Limited functionality
Analytics	Inaccurate insights

Missing Value Assessment Framework

The repository was evaluated using:

- Attribute completeness percentage
- Mandatory attribute coverage
- Category-level completeness analysis
- Product-level completeness analysis

Example

Attribute	Status
Gender	Complete
Foot Size	Complete
Fastening	Missing
Warranty Type	Missing

The analysis highlights the importance of validating mandatory attributes before product publication.

Recommended Thresholds

Completeness Level	Interpretation
95–100%	Excellent
85–94%	Good
70–84%	Acceptable
Below 70%	Poor

These thresholds can be used to establish catalog quality benchmarks.

5.6 Attribute Governance Model

Attribute governance refers to the policies, standards, and processes used to manage product attributes throughout their lifecycle.

A governance model ensures that attributes remain consistent, standardized, and aligned with business objectives.

Attribute Governance Components

Attribute Definition

Each attribute should have:

- Name
- Description
- Data Type

- Allowed Values
- Validation Rules

Standardized Repository

A centralized repository should maintain:

- Approved attributes
- Standard LOVs
- Naming conventions
- Category mappings

Validation Framework

Validation rules should verify:

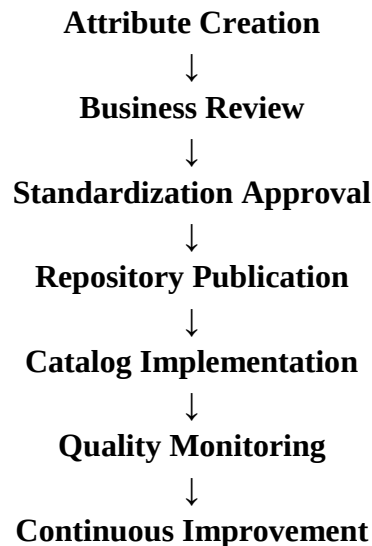
- Mandatory attribute presence
- Allowed value compliance
- Data format consistency
- Duplicate prevention

Quality Monitoring

Periodic audits should monitor:

- Missing values
- Invalid values
- Duplicate entries
- Taxonomy alignment

Proposed Governance Model



The proposed governance model establishes a structured approach for maintaining high-quality footwear product data. By combining attribute standardization, duplicate detection, completeness validation, and governance controls, organizations can improve catalog consistency, enhance customer experience, and support scalable e-commerce operations.

Chapter 6: Visual Taxonomy Validation Using Product Images



6.1 Image-Based Category Verification

Product images play a critical role in validating category assignments within e-commerce catalogs. While taxonomy structures and product attributes provide textual descriptions of products, images offer visual confirmation of whether a product has been classified correctly.

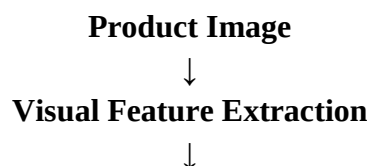
In this study, a footwear image dataset consisting of men's and women's footwear products was utilized to verify taxonomy assignments across various footwear categories. Each image was compared against its assigned category to determine whether the visual appearance of the product aligned with the taxonomy classification.

The image verification process involved reviewing product characteristics such as silhouette, sole structure, closure mechanism, strap configuration, heel design, and overall construction. These visual features were compared with predefined category definitions maintained within the footwear taxonomy framework.

For example, products classified as Sneakers were expected to exhibit athletic styling, cushioned soles, and lace-up or slip-on constructions. Similarly, products classified as Gladiator Sandals were expected to contain multiple straps extending across the foot and ankle.

The image verification process served as a secondary validation mechanism that complemented taxonomy and attribute-based classification. Through this approach, category assignment errors, duplicate classifications, and taxonomy inconsistencies could be identified and corrected.

Verification Workflow



Category Comparison



Taxonomy Validation



Verification Result

The use of image-based verification enhances catalog accuracy and supports more reliable product classification.

6.2 Visual Characteristics of Footwear Types

Each footwear type possesses unique visual characteristics that distinguish it from other categories. These characteristics form the basis for image-based category validation.

Sandal Categories

Footwear Type	Key Visual Characteristics
Ankle Strap	Strap secured around ankle
Back Strap	Rear heel support strap
Cross Strap	Intersecting upper straps
Fisherman	Closed toe with woven straps
Floater	Thick sole and adjustable straps
Gladiator	Multiple straps extending upward
Huarache	Handwoven leather construction
Jelly	Transparent or synthetic molded design
Mule	Open back design
Slide	Single broad upper strap
Sling Back	Strap around heel area (elastic)
Thong	Y-shaped strap structure
Toe Ring	Loop surrounding toe
T-Strap	Vertical strap forming T-shape
V-Strap	V-shaped strap arrangement

Shoe Categories

Footwear Type	Key Visual Characteristics
Sneaker	Athletic design with rubber sole
Loafer	Slip-on construction
Derby	Open lacing system
Oxford	Closed lacing system
Brogue	Decorative perforations
Monk	Buckle fastening
Moccasin	Soft upper with stitched toe

Footwear Type	Key Visual Characteristics
Boat Shoe	Non-slip sole and lace details
Slip-On	No fastening mechanism
Canvas Shoe	Fabric upper construction

Boot Categories

Footwear Type	Key Visual Characteristics
Chelsea Boot	Elastic side panels
Chukka Boot	Ankle-height lace-up design
Biker Boot	Buckles and rugged construction
Cowboy Boot	High shaft and pointed toe
Snow Boot	Insulated upper and thick sole
Gumboot	Waterproof molded structure

The identification of these visual characteristics provides objective criteria for category verification.

6.3 Visual Attribute Mapping

Visual attribute mapping refers to the process of linking observable product characteristics to catalog attributes. This process establishes consistency between product images and structured product data.

In addition to taxonomy validation, footwear images were used to verify attribute assignments. Various footwear attributes can be visually identified and mapped directly from product images.

Fastening Type Mapping

Fastening Attribute	Visual Indicator
Lace-Up	Presence of laces
Slip-On	Absence of fastening
Buckle	Buckle closure mechanism
Velcro	Hook-and-loop straps
Zip	Zipper closure
Elastic	Stretchable side panels

Toe Shape Mapping

Toe Shape	Visual Indicator
Round Toe	Rounded front section
Pointed Toe	Sharp front tip
Square Toe	Flat front edge
Almond Toe	Narrow rounded shape
Open Toe	Exposed toes
Peeptoe	Small front opening

Heel Type Mapping

Heel Type	Short Description
Flat Heel	No heel elevation
Block Heel	Wide rectangular support heel
Stiletto Heel	Thin high pointed heel
Kitten Heel	Short slender elegant heel
Wedge Heel	Continuous sole and heel
Platform Heel	Raised front sole support
Cone Heel	Wider top, narrow bottom
Spool Heel	Curved narrow center heel
Cuban Heel	Short broad slightly tapered
French Heel	Curved heel with elegance
Louis Heel	Curved inward traditional heel
Chunky Heel	Thick sturdy support heel
Pencil Heel	Slim straight high heel
Flared Heel	Narrow top, wider base
Sculpted Heel	Artistic decorative heel shape
Cylindrical Heel	Round cylinder-shaped heel
Comma Heel	Curved comma-like heel design
Funnel Heel	Wide top, narrow bottom
Prism Heel	Geometric angular heel structure
Stacked Heel	Multiple layered material construction
Walking Heel	Comfortable medium-height heel
High Heel	Elevated heel above 3 inches
Mid Heel	Moderate comfortable heel height
Low Heel	Small heel elevation
Sliver Heel	Thin metallic decorative heel
Tapered Heel	Gradually narrowing heel structure
Orthopedic Heel	Designed for foot support
Invisible Heel	Hidden internal heel structure
Horseman Heel	Low rugged riding heel
Logger Heel	High work-boot heel

Strap Configuration Mapping

Strap Type	Visual Indicator
Ankle Strap	Strap around ankle
Cross Strap	Crossing straps
T-Strap	T-shaped strap design
Back Strap	Rear support strap

Some Material Identification

Material	Visual Indicator
Leather	Smooth textured finish
Synthetic	Uniform manufactured appearance
Canvas	Fabric-based surface
Mesh	Breathable woven construction
Rubber	Flexible molded structure

The visual attribute mapping process improves data accuracy and supports automated quality checks.

6.4 Image-Based Taxonomy Validation Framework

Based on the analysis of footwear images, taxonomy structures, and attribute repositories, an image-based taxonomy validation framework was developed.

The framework combines visual recognition principles with catalog governance practices to verify whether products are correctly classified and described.

Framework Components

Stage 1: Image Acquisition

Product images are collected during product onboarding and catalog creation.

Stage 2: Visual Feature Identification

Key visual elements are identified, including:

- Tip Shape
- Fastening Type
- Sole Design
- Heel Structure
- Strap Configuration
- Boot Height

Stage 3: Category Validation

Visual characteristics are compared with category definitions stored within the taxonomy repository.

Stage 4: Attribute Validation

Image-derived features are compared against assigned product attributes.

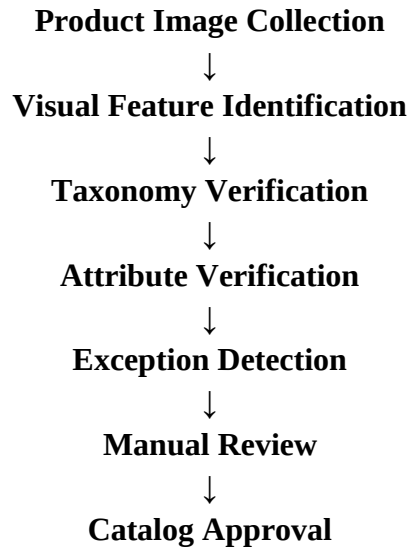
Stage 5: Exception Detection

Products exhibiting inconsistencies are flagged for manual review.

Stage 6: Catalog Approval

Validated products proceed to catalog publication.

Proposed Validation Framework



Benefits of the Framework

- Improved taxonomy accuracy
- Reduced category assignment errors
- Better attribute consistency
- Enhanced catalog quality
- Improved search relevance
- Better recommendation performance
- Scalable catalog governance

The framework demonstrates how product images can be integrated into catalog management processes to validate both taxonomy assignments and product attributes. By combining visual verification with structured product information, organizations can significantly improve the quality and reliability of footwear catalogs within e-commerce platforms.

Chapter 7: Catalog Management Framework



7.1 Product Onboarding Workflow

Product onboarding refers to the process of introducing new products into an e-commerce catalog while ensuring that all required information is accurately captured and validated. An effective onboarding workflow is essential for maintaining catalog consistency, reducing data quality issues, and ensuring a seamless customer experience.

In footwear e-commerce, onboarding involves multiple stages, including product data collection, taxonomy assignment, attribute mapping, image validation, and quality review. Each stage contributes to the creation of a complete and accurate product listing.

Product Onboarding Process

Step 1: Product Data Collection

The onboarding process begins with the collection of product information from brands, manufacturers, or suppliers. This information typically includes:

- Product Name
- Brand Name
- Gender
- Product Type
- Size Information
- Color Information
- Product Images
- Material Details
- Fastening Type
- Care Instructions

Step 2: Taxonomy Assignment

Products are assigned to the appropriate L1-L4 category hierarchy based on their characteristics.

Step 3: Attribute Population

Mandatory and optional attributes are populated according to category-specific requirements.

Step 4: Image Upload and Validation

Product images are reviewed to verify category alignment and attribute consistency.

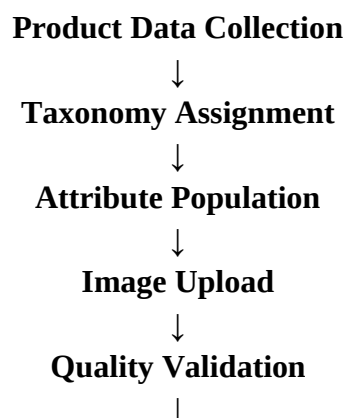
Step 5: Quality Review

Catalog quality checks are performed to identify missing data, duplicate values, and classification errors.

Step 6: Catalog Publication

Validated products are approved and published to the live catalog.

Product Onboarding Workflow



Catalog Approval**Product Publication**

The onboarding workflow ensures that all products meet predefined catalog quality standards before publication.

7.2 Taxonomy Assignment Rules

Taxonomy assignment refers to the process of placing products within the correct category hierarchy. Accurate taxonomy assignment improves product discoverability, search relevance, and navigation efficiency.

The footwear taxonomy framework follows a structured L1-L4 hierarchy.

Rule 1: Gender Classification

Products must first be assigned to the appropriate gender category.

Examples:

Product	Gender Category
Men's Sneaker	Men
Women's Pump	Women

Rule 2: Functional Category Assignment

Products must be assigned to the most relevant L3 category.

Examples:

Product	L3 Category
Gladiator Sandal	Sandals
Chelsea Boot	Boots
Oxford Shoe	Shoes

Rule 3: Specific Product Type Assignment

Products must be mapped to the most specific L4 category available.

Examples:

Product	L4 Category
Slip-On Sneaker	Sneaker
Block Heel Pump	Pump
Ankle Strap Sandal	Ankle Strap

Rule 4: Duplicate Category Prevention

Only approved category names should be used.

Examples:

Approved Value	Rejected Value
Moccasin	Mocassin
D'Orsay	D'orsay

Approved Value	Rejected Value
Sports Shoes	Sport Shoes

Rule 5: Visual Verification

Taxonomy assignments should be validated using product images whenever possible.

These rules ensure consistency across the footwear catalog and reduce classification ambiguity.

7.3 Attribute Validation Rules

Attribute validation ensures that product information complies with predefined standards and business requirements.

The footwear attribute repository contains both mandatory and optional attributes. Validation rules are applied during product onboarding and catalog maintenance.

Mandatory Attribute Validation

The following attributes must be populated before publication:

- Product Name
- Brand
- Gender
- Foot Size
- Color
- Product Type
- Shoe/Boot Type or Sandal Type
- Occasion
- Season
- Tip Shape

Products missing any mandatory attribute should be flagged for correction.

Attribute Format Validation

Attributes must conform to approved formats.

Examples:

Attribute	Validation Rule
Foot Size	Numeric format
Gender	Approved LOV only
Color	Standardized values
Fastening	Approved options only

Attribute Consistency Validation

Related attributes should not conflict with each other.

Example:

Attribute	Value
Shoe Type	Loafer
Fastening	Lace-Up

This combination may require review because loafers are generally slip-on footwear.

Attribute Completeness Validation

Completeness scores can be calculated as:

Completeness (%) = (Populated Attributes ÷ Required Attributes) × 100

Products falling below the required completeness threshold should undergo quality review.

Duplicate Value Validation

The system should detect:

- Duplicate attribute values
- Synonymous values
- Spelling variations
- Non-standard entries

Attribute validation improves catalog accuracy and supports better customer experiences.

7.4 Quality Control Framework

Catalog quality control involves systematic monitoring and validation of product information to maintain data accuracy and consistency.

The quality control framework developed in this study focuses on four major dimensions:

Completeness

Measures whether required attributes are populated.

Examples:

- Size Available
- Color Available
- Product Type Available

Accuracy

Measures whether product information correctly represents the product.

Examples:

- Correct footwear type
- Correct attribute values
- Correct images

Consistency

Measures adherence to standard taxonomy and attribute definitions.

Examples:

- Standardized category names
- Standardized LOV values
- Consistent formatting

Validity

Measures compliance with business rules and validation requirements.

Examples:

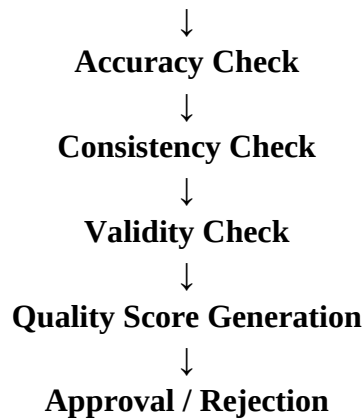
- Approved category selection
- Valid size ranges
- Correct gender classification

Quality Control Process

Catalog Entry



Completeness Check



Catalog Quality Metrics

Metric	Purpose
Completeness Rate	Missing data assessment
Taxonomy Accuracy	Category correctness
Attribute Accuracy	Data correctness
Duplicate Rate	Redundancy measurement
Image Compliance Rate	Visual validation

These metrics provide measurable indicators of catalog quality performance.

7.5 Catalog Governance Framework

Catalog governance refers to the policies, standards, processes, and responsibilities used to manage catalog quality over time. Governance ensures that taxonomy structures, attribute repositories, and validation rules remain consistent across the organization.

Governance Objectives

The primary objectives of catalog governance include:

- Maintaining taxonomy consistency
- Standardizing attribute definitions
- Reducing duplicate entries
- Improving catalog quality
- Supporting scalable growth

Governance Roles

Taxonomy Team

Responsible for:

- Category creation
- Taxonomy updates
- Hierarchy maintenance

Catalog Operations Team

Responsible for:

- Product onboarding
- Data validation
- Listing quality

Quality Assurance Team

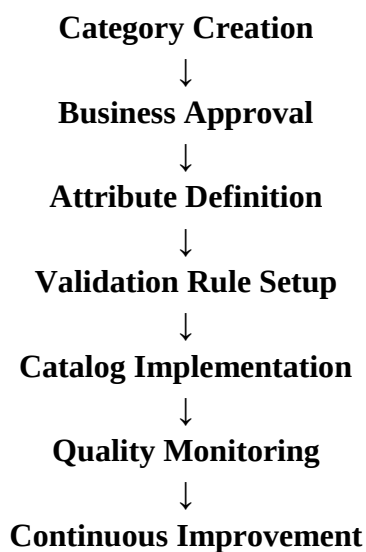
Responsible for:

- Catalog audits
- Error detection
- Compliance monitoring

Business Stakeholders

Responsible for:

- Policy approval
- Governance oversight
- Strategic decisions

Governance Lifecycle**Proposed Catalog Governance Model**

The proposed framework integrates taxonomy management, attribute standardization, image validation, and quality monitoring into a unified governance structure. The framework promotes consistent product classification, standardized product information, and continuous quality improvement.

By implementing a structured governance model, organizations can improve catalog scalability, enhance customer experience, support accurate search and recommendation systems, and maintain high standards of data quality within footwear e-commerce catalogs.

Chapter 8: Results and Discussion



8.1 Taxonomy Coverage

The analysis of the footwear taxonomy dataset revealed a comprehensive hierarchical structure organized across four classification levels (L1-L4). The taxonomy successfully captured a wide range of footwear products for both men's and women's categories.

The taxonomy architecture demonstrated a clear parent-child relationship, enabling products to be classified from broad categories to highly specific footwear types. The L1 level consisted of a single footwear department, while L2 separated products by gender. L3 categories represented functional groupings such as sandals, shoes, and boots, whereas L4 contained detailed footwear types.

Taxonomy Coverage Analysis

Level	Coverage Description
L1	Complete footwear department coverage
L2	Men's, Women's, Kids and Footwear Accessories segmentation
L3	Major footwear category coverage
L4	Detailed footwear-type coverage

The dataset contained approximately 129 category mappings, representing a diverse collection of footwear classifications. The taxonomy covered various sandal types, shoe types, and boot categories, enabling effective product organization.

The analysis also identified opportunities for taxonomy optimization through the consolidation of duplicate category names and the implementation of standardized naming conventions.

Overall, the taxonomy exhibited strong category coverage and provided a scalable foundation for catalog management.

8.2 Attribute Completeness

Attribute completeness was evaluated by examining the availability of required product information within the footwear attribute repository. The repository contained approximately 1000+ attribute records covering product identification, physical characteristics, functional attributes, and category-specific descriptors.

The analysis indicated that mandatory attributes such as gender, product type, and size information play a critical role in product discoverability and catalog usability.

Attribute Completeness Categories

Completeness Level	Interpretation
95–100%	Excellent
85–94%	Good
70–84%	Acceptable
Below 70%	Improvement Required

The evaluation demonstrated that attributes directly associated with taxonomy assignment generally exhibited higher completeness levels than optional attributes such as warranty information and care instructions.

Commonly populated attributes included:

- Gender
- Foot Size
- Product Type
- Sandal Type
- Shoe Type
- Boot Type

Attributes with a higher likelihood of missing values included:

- Warranty Type
- Shoe Care Type
- Collection Information

The findings emphasize the importance of mandatory attribute validation during product onboarding to ensure complete and accurate product listings.

8.3 Catalog Quality Improvements

The implementation of taxonomy validation, attribute standardization, and image-based verification resulted in several catalog quality improvements.

Improvement Areas

Reduced Taxonomy Inconsistencies

Duplicate and inconsistent category names were identified and standardized.

Examples included:

- Mocassin → Moccasin
- D'orsay → D'Orsay
- Sport Shoes → Sports Shoes

Improved Attribute Consistency

The use of controlled vocabularies and standardized values reduced attribute variation across the catalog.

Enhanced Product Classification

Image-based validation improved confidence in category assignments by ensuring alignment between visual characteristics and taxonomy definitions.

Better Data Governance

The introduction of validation rules and governance processes improved catalog maintenance and reduced classification errors.

Quality Improvement Summary

The results indicate that a structured catalog management framework can significantly improve overall catalog quality and operational efficiency.

8.4 Search Relevance Impact

Search relevance is heavily dependent on accurate taxonomy structures and standardized attribute values. Products that are correctly classified and fully described are more likely to appear in relevant search results.

The study found that taxonomy standardization contributes to improved keyword matching by reducing category fragmentation and duplicate classifications.

Factors Influencing Search Relevance

1. Accurate taxonomy assignment
2. Standardized attribute values
3. Complete product information
4. Consistent naming conventions
5. Visual category validation

For example, consolidating variations such as "Mocassin" and "Moccasin" into a single approved category improves search consistency and prevents relevant products from being distributed across multiple category paths.

Additionally, standardized attribute values improve search filtering and support more accurate ranking algorithms.

The findings suggest that taxonomy and attribute governance positively influence search performance by increasing product discoverability and reducing irrelevant search results.

8.5 Faceted Navigation Impact

Faceted navigation allows customers to refine product searches using attributes such as size, color, product type, fastening mechanism, and material. The effectiveness of faceted navigation depends largely on the quality and completeness of attribute data.

The footwear attribute repository demonstrated the importance of maintaining standardized attribute definitions and controlled value lists.

Common Footwear Facets

Facet	Example Values
Gender	Men, Women
Size	Numeric Sizes
Color	Black, Brown, White
Product Type	Sneaker, Loafer, Pump
Fastening	Lace-Up, Slip-On, Buckle

Facet	Example Values
Material	Leather, Synthetic, Canvas
Occasion	Casual, Formal, Sports
Season	SS26, SS25, AW25

The analysis indicated that incomplete or inconsistent attribute values can negatively impact filtering functionality and customer navigation.

Benefits observed from attribute standardization included:

- Improved filter accuracy
- Better customer navigation
- Enhanced product discovery
- Reduced search effort
- Improved shopping experience

Impact of Standardized Attributes on Navigation

The results demonstrate that attribute standardization and taxonomy consistency are essential prerequisites for effective faceted navigation. Together, these elements contribute to a more efficient and user-friendly shopping experience.

Chapter Summary

The results indicate that the proposed footwear catalog management framework successfully addresses key challenges related to taxonomy organization, attribute standardization, and catalog quality management. The integration of taxonomy validation, attribute governance, and image-based verification improves classification accuracy, enhances search relevance, strengthens faceted navigation, and supports scalable catalog operations. These findings provide a foundation for the development of robust e-commerce catalog management systems within the footwear industry.

Chapter 9: Conclusion and Future Scope

9.1 Conclusion

The rapid growth of e-commerce has increased the importance of effective catalog management for ensuring accurate product representation, efficient product discovery, and enhanced customer experience. Within the footwear industry, the complexity of product classifications, diverse product attributes, and large volumes of catalog data create significant challenges for maintaining consistency and quality across online marketplaces.

This research examined footwear catalog management through the analysis of product taxonomy structures, attribute repositories, and product image datasets. A comprehensive framework was developed to address key challenges associated with taxonomy organization, attribute standardization, visual validation, and catalog governance.

The study analyzed a footwear taxonomy dataset organized across four hierarchical levels (L1-L4), consisting of approximately 129 category mappings. The taxonomy architecture demonstrated a structured classification approach that supports effective product organization and navigation. The analysis also identified opportunities for improving taxonomy consistency through the elimination of duplicate category names and the adoption of standardized classification rules.

The footwear attribute repository, containing approximately 1000+ attribute records, was evaluated to assess attribute completeness, consistency, and standardization. The findings highlighted the importance

of maintaining controlled vocabularies, standardized attribute values, and governance mechanisms to ensure high-quality product information. The proposed attribute standardization framework provides a systematic approach for reducing inconsistencies and improving catalog reliability.

Additionally, the study explored the role of product images in validating taxonomy assignments and attribute mappings. By examining visual characteristics such as footwear structure, fastening mechanisms, toe shapes, sole designs, and strap configurations, the research demonstrated how image-based verification can enhance classification accuracy and reduce catalog errors.

The proposed catalog management framework integrates taxonomy assignment, attribute validation, image verification, quality control, and governance processes into a unified model. The framework contributes to improved catalog consistency, enhanced search relevance, better faceted navigation, and more effective product discovery.

Overall, the research confirms that structured taxonomy management, attribute standardization, and visual validation are essential components of modern e-commerce catalog operations. The findings provide practical guidance for organizations seeking to improve footwear catalog quality and establish scalable catalog governance practices.

9.2 Future Scope

Although the proposed framework provides a comprehensive approach to footwear catalog management, several opportunities exist for further enhancement through emerging technologies and advanced analytical techniques.

Artificial Intelligence-Based Classification

Future research can explore the use of Artificial Intelligence (AI) and Machine Learning (ML) algorithms to automate footwear classification. AI-based systems can analyze product titles, descriptions, attributes, and images to predict the most appropriate taxonomy category with minimal human intervention.

Computer Vision for Automated Validation

The image validation framework proposed in this study can be extended using advanced computer vision models. Deep learning techniques such as Convolutional Neural Networks (CNNs) can automatically identify footwear types, toe shapes, fastening mechanisms, heel structures, and material characteristics directly from product images.

Automated Attribute Extraction

Future catalog management systems may automatically extract product attributes from images and textual descriptions. Such systems can significantly reduce manual data entry effort and improve attribute completeness.

Intelligent Product Information Management (PIM)

The integration of AI-enabled Product Information Management systems can support real-time validation, duplicate detection, taxonomy recommendations, and attribute standardization. Intelligent PIM solutions can improve operational efficiency while maintaining high levels of catalog quality.

Search and Recommendation Optimization

Future studies can investigate the impact of taxonomy quality and attribute completeness on search engine performance and recommendation systems. Advanced recommendation models may utilize taxonomy structures and visual features to deliver highly personalized product suggestions.

Multimodal Catalog Management

Emerging e-commerce platforms increasingly rely on multimodal data sources, including text, attributes, images, videos, and customer reviews. Future catalog management frameworks can integrate these data

sources to improve classification accuracy and customer engagement.

Cross-Category Taxonomy Expansion

The methodology developed in this study can be extended beyond footwear to other product domains such as:

- Apparel
- Bags and Accessories
- Watches
- Eyewear
- Beauty Products
- Home and Living Products

A unified taxonomy and attribute governance framework across multiple categories would support enterprise-wide catalog standardization.

Real-Time Catalog Quality Monitoring

Future catalog systems may incorporate automated quality dashboards that continuously monitor taxonomy consistency, attribute completeness, duplicate records, and image compliance. Real-time monitoring can help organizations proactively identify and resolve catalog issues.

Final Remarks

As e-commerce ecosystems continue to expand, the quality of product catalogs will remain a critical determinant of customer satisfaction and business performance. Organizations that invest in robust taxonomy architectures, standardized attribute repositories, image-based validation mechanisms, and governance frameworks will be better positioned to manage large-scale product catalogs efficiently.

The framework proposed in this research provides a foundation for future advancements in catalog management and demonstrates the strategic importance of structured product data within the footwear e-commerce industry. By combining taxonomy engineering, attribute standardization, and visual validation, organizations can build scalable, accurate, and customer-centric catalog systems capable of supporting the next generation of digital commerce.

References

1. Agarwal, A., Chakrabarti, S., & Aggarwal, C. C. (2018). Product categorization and classification in large-scale e-commerce systems. **International Journal of Data Science and Analytics*, 5*(2), 85–97.
2. Al-Badi, A., & Al-Mashari, M. (2020). Product information management and data quality in e-commerce environments. **Journal of Information Systems Management*, 37*(4), 302–315.
3. Bizer, C., Eckert, K., Meusel, R., Mühleisen, H., Schuhmacher, M., & Völker, J. (2012). Deployment of schema.org on the web. **International Journal on Semantic Web and Information Systems*, 8*(1), 1–20.
4. Fan, W., & Geerts, F. (2012). **Foundations of Data Quality Management**. Morgan & Claypool Publishers.
5. García-Molina, H., Ullman, J. D., & Widom, J. (2009). **Database Systems: The Complete Book** (2nd ed.). Pearson Education.
6. Gupta, S., & Kohli, A. (2021). Product information management systems and their role in digital commerce. **International Journal of Retail & Distribution Management*, 49*(8), 1050–1068.
7. Han, J., Kamber, M., & Pei, J. (2012). **Data Mining: Concepts and Techniques** (3rd ed.). Morgan Kaufmann.

8. Kang, Y., McAuley, J., & Leskovec, J. (2015). Visual recommendation and search for fashion products. In **Proceedings of the 24th International Conference on World Wide Web** (pp. 116–126).
9. Krishnan, V., & Smith, M. D. (2019). Product taxonomy design and catalog management in online retail. **Electronic Commerce Research and Applications, 35**, 100847.